

Thermal Quadrupoles

Solving the Heat Equation through
Integral Transforms

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Preface

The material and the methods that are presented in this book have been developed in our research laboratory since the start of the 1980s. The result should be considered as a collective work.

Originally the quadrupole method was developed as a tool for developing new techniques of thermal property measurement. In this kind of situation an experiment involving heat transfer has first to be designed. The sample, whose thermophysical properties are looked for (diffusivity, conductivity, etc.), is subjected to a thermal excitation and different data (thermocouple potentials, electrical heating, etc.) are recorded. Comparison of these data with the output of a thermal model allows, through least squares minimization, for example, estimation of its thermal parameters. The quadrupole method allows the construction in a very simple way, of an exact and reliable direct model that can be used in the optimal design of the experiment as well as in the 'best fit' inversion process.

Very rapidly it became apparent that this method should not be restricted to research applications. It has therefore been taught within the framework of senior and post-graduate classes at ENSEM (an engineering school of the 'Institut National polytechnique de Lorraine', in Nancy) and later in other engineering schools such as INSA ('Institut National des Sciences Appliquées', in Lyon) or ENSAM ('Ecole Nationale Supérieures des Arts et Métiers', in Bordeaux).

From a mathematical point of view, the quadrupole method belongs to the class of analytical methods for solving linear partial differential equations in simple geometries. It relies on classical analytical tools such as Laplace integral transforms (in time) or space integral transforms (Fourier, Hankel) related to the method of separation of variables. Applied to the heat equation, the thermal 'solution' can be expressed in terms of linear matrix relationships between transformed temperature flux vectors at the boundaries of the system. This allows intrinsic solutions to be obtained whose form does not depend on these boundary conditions. This approach is common to transient or steady-state situations, and to one- or multidimensional transfers. It can also be extended to periodic regimes.

Solving the heat equations through integral transforms does not constitute a new subject. However, this kind of traditional analytical method is now supplemented by modern mathematical methods such as inverse Laplace transforms or inverse fast Fourier transforms. These numerical techniques can be implemented through the use of powerful software packages such as MATLAB[®]. This gives rise to efficient hybrid numerical-analytical constructions for the temperature field, constructions that were not available 20 years ago. In this manner we hope that our book can bridge the gap between numerical and analytical methods. Although often very powerful, numerical methods require a space gridding that can be implemented in very different ways since it is not unique. The resulting numerical process puts some distance between the solution and the data because this solution cannot be related to the data in a formal way. Analytical

methods, which are usually considered as limited in application (very often only homogeneous materials can be treated, which is not the case for our method), have the advantage of bringing a physical insight and a 'transparency' to the problem at stake, because of their explicit nature.

As any engineer may have learned it, finding a solution generally constitutes only one part of the problem. In design problems an initial thermal design has to be tested through the calculation of the temperature or flux field, followed by an analysis of this field in terms of constraints. A modified design is then proposed, followed by a new thermal field calculation, and so on until the right design is found. The thermal quadrupole method allows this often painful iterative procedure to be removed by allowing only one calculation that can be done advisedly: a network representation that relies on the notion of explicit 'thermal impedances' often allows a reduction, i.e. a simplification, of the first tested model. This explicit analytical knowledge of the different impedances allows us to come straight to the point: even if this simplification corresponds to a loss of accuracy in the temperature or heat flux solution, it brings a quick response to many qualitative questions (the sensitivity of the design to one parameter, the one- or multidimensional character of the transfer, etc.). This impedance representation leads directly to a systemic modelling associated with the construction of transfer functions of complex thermal systems.

We hope this book will become a useful reference for professionals and engineers involved in heat transfer measurements and thermal characterization. It can also be of some interest to people needing a sound basis for comparison with their numerical codes, if they do not want to invest in too-complicated analytical methods. Since our method is analytical, it can also be used as a benchmark for people interested in the development of numerical methods. It can be useful for process engineers needing to model a heat transfer problem in order to obtain optimal operating conditions. Researchers interested in simulation or design of experiments where heat transfer plays a significant role may find practical responses to their modelling problem in this book. Other types of transfers, such as petroleum flow in a porous reservoir, even if it is not dealt with in the book, can also be solved by a quadrupole technique directly derived from thermal quadrupoles (the Fourier law has to be replaced by Darcy's law); this type of method is now being used by petroleum engineers. Fans of matrix solvers such as MATLAB[®] can also find new fields of application in the material presented in this book.

Our method is quite efficient for solving inverse thermal problems: integral transform methods are well-suited for inverting measurements produced by modern data acquisition systems (numerical oscilloscopes, infrared cameras etc.). One can mention, at this level the problems of wall heat flux estimation from internal temperature measurement or of characterization of delaminations in composite materials from infrared camera measurements, etc.

The chapters follow an order of increasing complexity: the quadrupole method is first presented (Chapter 1) and the different notions of conduction in simple geometries are recalled (Chapter 2). The 'heart' of the method is presented in transient one-dimension heat transfer (Chapter 3) and an extension to multidimensional cases is developed next (Chapter 4). The same approach can be used for time-periodic regimes (Chapter 5). Advanced techniques, based on the same philosophy but often requiring more intensive numerical implementation, are presented in the next chapter (Chapter 6). The quadrupole method is extended then to problems that are not based on the conduction equation:

dispersion of a solute in a flow of solvent in a porous medium (mass dispersion, see Chapter 7) and simultaneous conduction and radiation heat transfer in a semi-transparent medium (Chapter 8). The final chapter is devoted to the practical numerical methods that can be used to invert the Laplace transform (Chapter 9). A wide range of problems of contemporary interest are presented, mainly in Chapters 3 and 6, with their numerical implementation in MATLAB[®] language. We also provide solutions to these problems.

The material in this book can form the basis for a two semester course: Chapters 1–5 and Chapter 9 can be used within a first course in conduction heat transfer at the upper graduate level; Chapters 6–8 can be used in a more specialized course on diffusive and radiative transfers or in a thermal measurement course, at graduate level. This is the way it is used currently in Nancy.

The quadrupole method is not only a method but also a culture. It has been used by many colleagues who prepared their theses at LEMTA ('Laboratoire d'Energétique et de Mécanique Théorique et Appliquée') during the last 15 years, before migrating to new horizons in France or abroad. Each of them has also brought his brick to the construction presented in this book. We wish to thank in particular Mouloud Amazouz, Smail Azizi, Abdelhakim Bendada, Sophie Didierjean, Anne-Sophie Houllbert, Mohamed Lachi, Bruno Ladevie, Yannick Lescanne, David Mourand, Isabelle Philippi-Péry, Benjamin Rémy, Xinxin Zhang and Xiaojing Yin-Zhang.

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1

Interest of the Quadrupole Approach

1.1 THE QUADRUPOLE METHOD: USE, ORIGIN AND PRESENTATION IN THIS BOOK

We will try to explain in this section both the interest and the origin of the quadrupole method in relationship to the other chapters of this book.

The first reaction of the engineer who wants to model a diffusive heat transfer within a medium or within a solid structure is to use the main numerical tools and the corresponding codes that are available in the market: finite differences, finite elements, finite volumes, boundary elements and so on.

Very often, during his period of formation, he has come into contact with analytical modelling tools but, except in the very simple case of a steady-state, one-dimensional transfer in simple geometry where the use of the electrical analogy (thermal resistances) is very popular, he does not feel inclined to employ these types of tools.

The use of a numerical method is usual whenever the geometry is complex or the problem non-linear. In the opposite cases, an analytical method can be considered: the method of separation of variables allows solutions to be found that have the form of time-space series (in finite dimensions) and, under certain conditions, the method of Green's functions provides solutions in an integral form for the temperature field.

However, these methods are rather painful to implement in practical situations and simulation of heat transfer within a multimaterial such as a multilayer composite medium rapidly becomes a nightmare for the non-specialist (calculation and separation of eigenvalues, Green's functions with different forms according to the time or space regions, etc.).

Another technique exists: instead of looking for a solution that is a temperature in the usual time-space domain, it is possible to seek a solution in a transformed domain. So, if one deals with a one-dimensional transient problem, a Laplace transform on the time variable will produce a 'Laplace temperature' that will often exhibit a very simple explicit form instead of the series of the initial problem.

This approach is not a novel one; until recently it stumbled over the return into the original space either because of ignorance of efficient inversion methods or because of lack of computing power. For example, and to consider again the preceding example, one could hope to find the solution in the space domain only for a restricted class of functions in the Laplace domain, functions for which the original was known analytically (tables of Laplace transform pairs). Presently this difficulty can be overcome because a certain number of inversion tools exist, including numerical tools, for a very broad class of functions (see Chapter 9).

A very important qualitative step can be taken if, after Laplace transforming the precedent problem, a temperature-flux vector is considered at one point of the medium. It is therefore

possible to replace the heat equation and its associated initial condition by a linear matrix equation that links an input vector (Laplace temperature and flux at one boundary of the medium) to the corresponding output vector at the other boundary: the heat equation is not yet solved (boundary conditions are not yet given) but an intrinsic relationship between input and output of the medium becomes available. The writing of this equation for each layer of a multilayer composite slab provides the same kind of input/output relationship after elimination of the interface vectors. The complete solution will be found if two of the four 'poles' (input or output Laplace temperature or Laplace flux) are given (see the other sections of Chapter 1 for a brief overview of this method and its advantages stemming from its explicit analytical approach) and Chapter 3 for its extended presentation in one-dimensional cases.

This approach was first presented by H. S. Carslaw in his 1921 book (*Introduction to the Mathematical Theory of Heat in Solids*, Macmillan) for the complex harmonic temperature and flux representation (time-dependent periodical regimes where the formalism is the same as the one described above for transient regimes with the Laplace transformation) and by H. S. Carslaw and J. C. Jaeger in their 1946 and 1959 book (*Conduction of Heat in Solids*, first and second editions, Oxford University Press) for the Laplace transform representation (see p. 326, Chapter 12, of the second edition).

Carslaw and Jaeger did not give any precise name to this method but refer to the four terminal network method used in electrical circuits (a quite unattractive qualifier); since the above-mentioned equation relates four scalars or poles, it seems quite natural to call it the quadrupole method. This matrix approach can be modelled by the association of impedances that allows the use of 'thermal' impedances that are analogous to electrical impedances in an electrical transmission line. The treatment of a heat transfer problem then becomes very similar to the modelling of electrical circuits (once a boundary condition is known), the electrical current being replaced by the Laplace heat flux and the electrical potential by the Laplace temperature.

This systemic approach can be generalized to cases where the transfer is multidimensional (for simple geometries) with the use of appropriate integral transforms in one or several space variables; in any case, only one space direction remains untransformed in the quadrupole representation (see Chapters 4 and 6). The resulting representation brings a compactness to the solutions that are moreover explicit, which allows us to pinpoint the physical aspects of the considered problem. Moreover, a change of boundary conditions does not require one to rewrite the whole problem starting from scratch: the quadrupole relationship does not depend on the boundary conditions on the surfaces defined by the transformed directions.

The same type of technique allows the processing of more complex cases. The modelling of the transfer through an imperfect contact between two solids is considered in Chapter 6 (constriction resistances and impedances). The same chapter presents cases where the coefficients present in the boundary or interface conditions are no longer uniform or constant on the corresponding surfaces: such is the case for a space varying interface resistance or for an intermittent contact between two solids. An association of the quadrupole method to the finite difference method (hybrid approach) allows the modelling and the solution of transfer through a longitudinal two-layer material.

Another type of problem that can be tackled concerns the time-dependent periodic regimes (see Chapter 5). The formalism is the same: just replacing the Laplace parameter by the product of the time frequency and the imaginary number whose square is the negative

unit. In this kind of modelling the Laplace temperatures and fluxes are replaced by 'phasors', which are the complex amplitudes of temperatures and fluxes. This approach is quite natural to people accustomed to electrical circuits in a harmonic regime. In contrast to the Laplace transform representation, no numerical inversion is required in that case.

The quadrupole method is not solely reserved for heat conduction or diffusive transfers. Chapter 7 shows an application of this method to mass transfer in porous media, i.e. the transport of a tracer. This transport can be caused by simultaneous diffusion and convection.

Both simultaneous conductive and radiative transfers in semitransparent media can also be tackled by this method under certain assumptions (linearization of the radiative transfer equation for moderate temperature differences within the material and approximation of the kernels present in this equation). This is the subject of Chapter 8.

In order to present all these applications of the quadrupole method on a sound heat transfer basis, Chapter 2 is devoted to the main concepts required for solving the heat equation, with a strong emphasis on the different kinds of boundary and interface conditions. This chapter can be skipped by the reader familiar with heat conduction.

An extended list of references that covers the general subject of heat conduction, of Laplace transforms and their numerical inversion, of integral transforms, of heat transfer in multilayer materials and of the fundamentals and application of thermal quadrupoles is given at the end of this chapter. It allows the reader to grasp the present literature on the type of analytical methods that are presented throughout this book.

1.2 DEFINITION OF THE QUADRUPOLE METHOD

The quadrupole method is a unified exact explicit method of representation of linear systems. It can apply to a large number of classes of linear systems. We have decided to present it here within the framework of conductive transfer even though radiative heat transfer or mass diffusion examples are treated in two chapters of this book. Although there are several versions of this method, related to the type of problem at stake, it is based on a common methodology. In its simpler more basic version in heat conduction, it is based on 2×2 matrices that relate some transform of both temperature and flux on one surface of the considered medium to the same quantities on another surface. We use the term 'transform' here in a very wide meaning: it can be a Laplace transform in a transient one-dimensional case, a Fourier space cosine transform in a steady two-dimensional case, or a phasor (complex amplitude of temperature or flux) in a one-dimensional time-periodic case. It can also be a combination of several of the preceding transforms depending on the type of transfer that is considered. It allows the calculation of temperature and flux vector fields, but is not limited to this particular utilitarian aspect: it also brings a *representation* of heat transfer through multimaterials by networks of impedances similar to those used for the modelling of electrical circuits (transmission lines for example).

Basically, since its base is analytical, it is restricted to linear systems with simple geometries. These limits can however be alleviated: irregular geometries can be processed by the method, once a particular flux pipe can be defined between two surfaces, the considered temperatures becoming now average temperatures. Non-linear problems could also be treated if time remains untransformed: the basic technique can be extended by a numerical discretization of the time derivative. However, this extension does not enter the framework of this book since we have tried to focus primarily on the analytical aspects of the method.

1.3 HOW CAN IT BE WRITTEN?

In order to understand the way the quadrupole method can be implemented, two general heat transfer situations that are based on the notion of thermal resistance or of heat capacities will be briefly presented. They will be used as a base to introduce quadrupoles in a transient one-dimensional case.

1.3.1 Case a: Steady-State Transfer

Let us consider a medium made out of a single homogeneous material and limited by a closed surface of area S as depicted in Figure 1.1. We consider the steady-state heat transfer case where this medium receives heat from the outside through a part S_1 of surface S and releases it through another part S_2 . It is considered here that the remaining part of S is insulated from the outside environment. The incoming heat flux through S_1 is called Φ_1 while the outgoing flux through S_2 is called Φ_2 . We consider the average temperatures T_1 and T_2 over these two surfaces. Since we are under steady-state conditions, none of the four quantities T_1 , T_2 , Φ_1 and Φ_2 depends on time: all of them are constant. They are related by the two following equations:

$$T_1 - T_2 = R\Phi_1 \quad (1.1)$$

$$\Phi_1 = \Phi_2 \quad (1.2)$$

where R is the thermal resistance of the flux pipe between S_1 and S_2 . This resistance depends on the geometry of the medium and on its thermal conductivity λ .

The preceding two equations can be easily given a single matrix equation form:

$$\begin{bmatrix} T_1 \\ \Phi_1 \end{bmatrix} = \begin{bmatrix} 1 & R \\ 0 & 1 \end{bmatrix} \begin{bmatrix} T_2 \\ \Phi_2 \end{bmatrix} \quad (1.3)$$

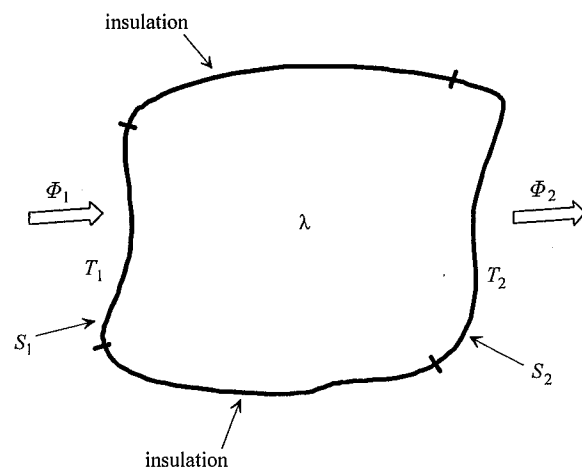


Figure 1.1 The steady-state case – the physical system

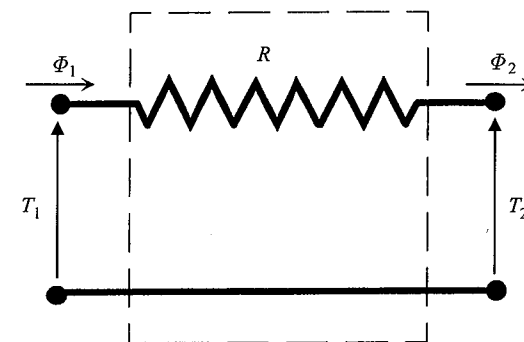


Figure 1.2 The steady-state case – network representation

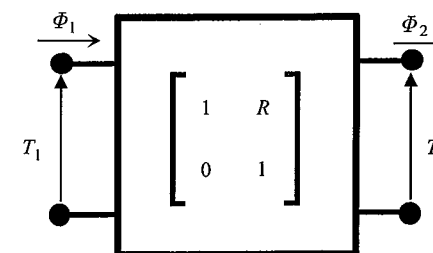


Figure 1.3 The steady-state case – matrix representation

This type of representation now uses the temperature vector $[T \ \Phi]^t$ as a whole on the two different surfaces (we use the 't' exponent here to denote the transpose of a matrix). In a similar way it is possible to set out an electrical representation of equation (1.3) with the use of a four-terminal network (also called a two-terminal pair network by electrical engineers) presented in Figure 1.2. In this representation one can observe a base line on the lower part of the figure: this simply indicates that the reference temperature is the same on the two sides of the system. There are four terminals in this representation (indicated by bold dots). The two heat fluxes Φ_1 and Φ_2 play the same part as electrical currents; they are represented here by horizontal arrows on the upper horizontal line. The two temperatures T_1 and T_2 are similar to the electrical potentials, which is why they are represented here as vertical arrows with respect to the base line (these vertical arrows stem from the mechanical analogy, where the potential of gravity is an altitude and natural movement of a body occurs in the direction of the slope).

An alternative representation of equation (1.3) is shown in Figure 1.3, where the constitutive network is replaced by the 2×2 matrix of equation (1.3).

It is interesting to note that either this matrix or the unique resistance of the network in Figure 1.2 completely characterizes the physical system.

1.3.2 Case b: A Lumped Body in Transient Conditions

The same physical system as above is now considered, but the transient conditions are now studied. The medium is initially at a uniform zero temperature and at time $t = 0$ it begins to

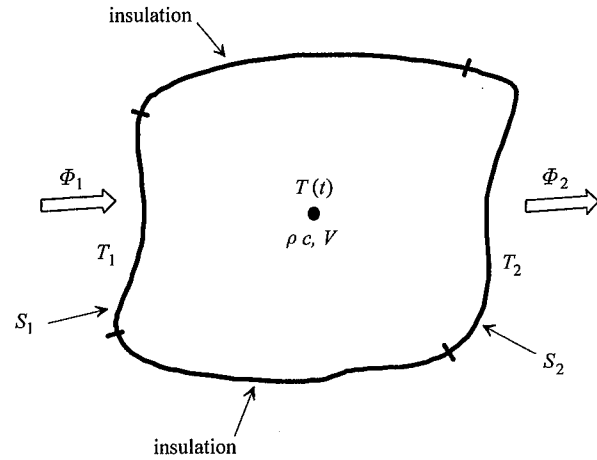


Figure 1.4 Lumped body model – physical system

exchange heat with the outside environment. It receives a heat flux $\Phi_1(t)$ through its surface S_1 and a flux $\Phi_2(t)$ through S_2 , the remaining part of S being insulated. Its conductivity λ is large enough to assume that at any time its temperature $T(t)$ can be taken as uniform, which means that it does not depend on the point considered. A simple heat balance during a time interval dt around t can be written as

$$\rho c V \frac{dT}{dt} = \Phi_1(t) - \Phi_2(t) \quad \text{with } T(0) = 0 \quad (1.4)$$

where ρ is the mass density, c the specific heat and V the volume of the medium (see Figure 1.4).

It is possible to remove the time derivative in equation (1.4) if the Laplace transforms $\theta(p)$ and $\phi_i(p)$ of the temperature and fluxes (for $i = 1$ or 2), respectively, are introduced:

$$\theta(p) = L[T(t)] = \int_0^\infty T(t) \exp(-pt) dt$$

and

$$\phi_i(p) = L[\Phi_i(t)] = \int_0^\infty \Phi_i(t) \exp(-pt) dt \quad (1.5)$$

Equations (1.4) and (1.5) then become

$$C_t p \theta(p) = \phi_1(p) - \phi_2(p) \quad (1.6)$$

where $C_t = \rho c V$ is the heat capacity of the medium. The temperatures $T_1(t)$ and $T_2(t)$ of the surfaces S_1 and S_2 are both equal to the temperature $T(t)$ in the medium since this temperature is uniform at any time t . The same is true for their Laplace transforms, i.e.

$$\theta(p) = \theta_1(p) = \theta_2(p) \quad (1.7)$$

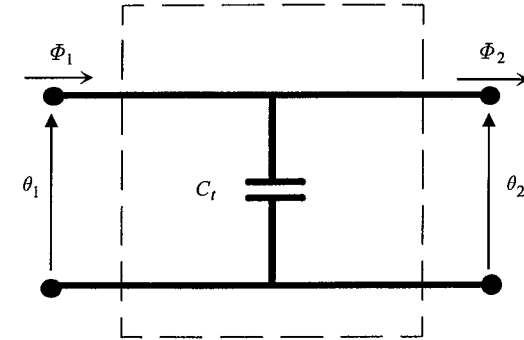


Figure 1.5 Lumped body model – network representation

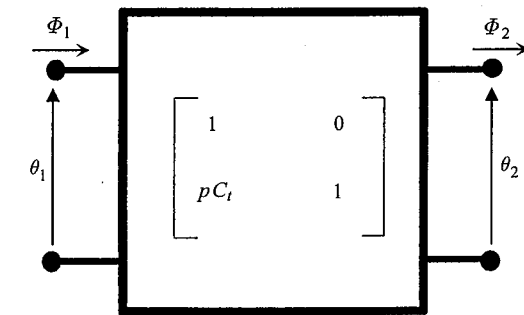


Figure 1.6 Lumped body model – matrix representation

Equations (1.7) and (1.6) give rise to the following matrix equation:

$$\begin{bmatrix} \theta_1 \\ \phi_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ C_t p & 1 \end{bmatrix} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} \quad (1.8)$$

This equation has exactly the same form as equation (1.3) in the steady state, but the temperature-flux vector $[T \ \Phi]^t$ at the non-adiabatic boundaries is replaced by its Laplace transform $[\theta \ \phi]^t$.

We have seen that in steady-state conditions the fluxes are equal, while the temperatures differ. For the lumped body assumption, the temperatures remain equal while the exchanged fluxes differ. By analogy with what is usually done in electricity, there is a four-terminal network representing this last situation (see Figure 1.5). It is also possible to use the matrix representation shown in Figure 1.6.

This matrix, or the unique heat capacity C_t , completely characterizes the thermal system for this model.

1.3.3 Case c: The General Case

The preceding presentation (cases a and b) were based on the decomposition of the boundary of the homogeneous medium into three parts: two active surfaces and an insulated one. The

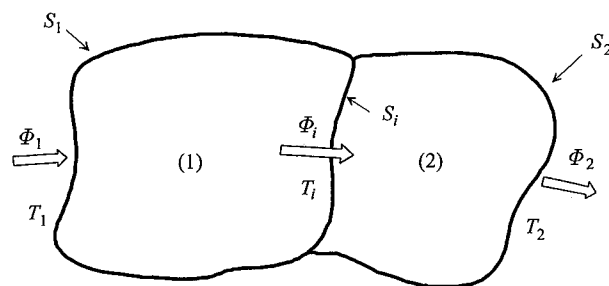


Figure 1.7 Cascade of two media

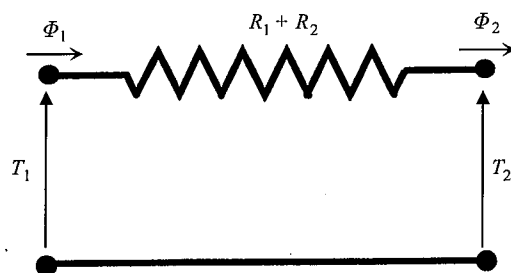


Figure 1.8 Cascade of two media - steady-state case

surface temperature-flux vector representation makes possible the modelling of multi-materials, i.e. systems composed of several different media that can be assembled in cascade for example. The case of a system composed of two different materials (1) and (2) is shown in Figure 1.7. If S_1 and S_2 are the outside surfaces of medium (1) and (2) and S_i their interface, then it is easy to represent the global transfer between S_1 and S_2 by a product of matrices.

With the same notation as above, the steady-state problem can be modelled in the following way:

$$\begin{bmatrix} T_1 \\ \Phi_1 \end{bmatrix} \begin{bmatrix} 1 & R_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & R_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} T_2 \\ \Phi_2 \end{bmatrix} = \begin{bmatrix} 1 & R_1 + R_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} T_2 \\ \Phi_2 \end{bmatrix} \quad (1.9)$$

A unique matrix, based on knowledge of the sum of the internal resistances of the two media $R_1 + R_2$, now links the temperature-flux vector, since the interface temperature vector $[T_i \ \Phi_i]^T$ can be eliminated if the two equations similar to (1.3) are written for each medium (between S_1 and S_i and between S_i and S_2). The corresponding network representation is shown in Figure 1.8.

The transient lumped body counterpart of this case can be derived in the same way:

$$\begin{bmatrix} \theta_1 \\ \phi_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ C_{11}p & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ C_{12}p & 1 \end{bmatrix} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ (C_{11} + C_{12})p & 1 \end{bmatrix} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} \quad (1.10)$$

It should be noted that the heat capacities of the individual media simply add up when the whole system is considered. Its network representation is shown in Figure 1.9.

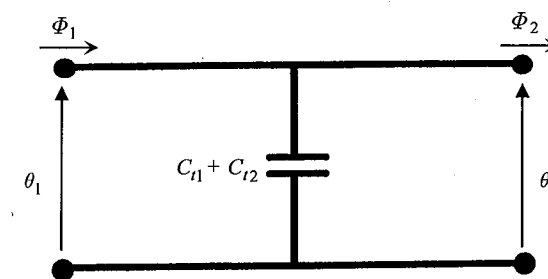


Figure 1.9 Cascade of two media - Lumped body approximation

The preceding two models (1.3) and (1.8) belong to the class of linear heat conduction models. The practical implementation of model (1.3) in the solution of a steady-state problem depends mainly on the way the internal resistance R can be evaluated. It also relies on the notion of average surface temperature, which means that a local temperature cannot be obtained. In that sense it is an external representation of the corresponding physical system, which links the four global variables (surface temperatures and fluxes) thanks to the definition of the thermal resistance of the corresponding flux pipe.

The application of the lumped body model requires the strong assumption of a uniform temperature distribution in the medium. It can be considered as a zero-dimensional model, since no geometrical data enter in the definition of the heat capacity that completely characterizes this system (it is simply the product of its mass by the specific heat c of the material).

The reader might wonder whether it is possible to extend models (1.3) and (1.8) to more general situations. For example, the multidimensional case shown in Figure 1.7 can be considered: transient conditions now apply and the temperature in each medium can no longer be taken as uniform. Since the problem is transient and linear, it is possible to write it in the Laplace domain and a relationship of the following form can be derived:

$$\begin{bmatrix} \theta_1 \\ \phi_1 \end{bmatrix} = \begin{bmatrix} A(p) & B(p) \\ C(p) & D(p) \end{bmatrix} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} = \mathbf{M}(p) \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} \quad (1.11)$$

A question may then arise: Can the preceding matrix representations be used to characterize completely the matrix $\mathbf{M}$ using knowledge of the thermal resistance and the heat capacity of each medium? For example, is it legitimate to write

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ C_{11}p & 1 \end{bmatrix} \begin{bmatrix} 1 & R_2 \\ 0 & 1 \end{bmatrix} \quad \text{or} \quad \mathbf{M} = \begin{bmatrix} 1 & R_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ C_{12}p & 1 \end{bmatrix} ? \quad (1.12)$$

The answer is no if no further assumption is made about the nature of the two media. The coefficients of the $\mathbf{M}$ matrix cannot be derived, in the general case, by a simple multiplication of resistive or capacitive matrices.

However, things simplify a lot in the case of a one-direction heat transfer.

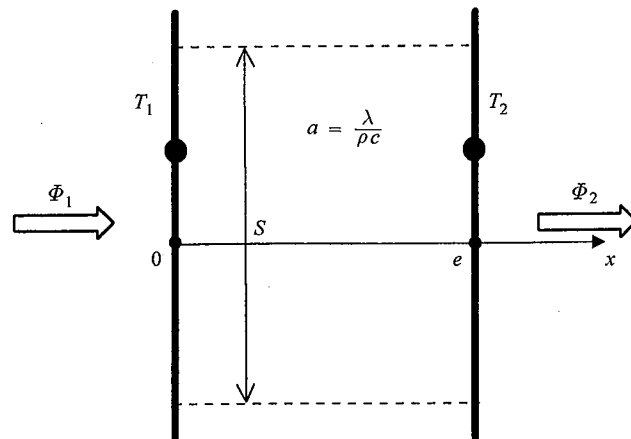


Figure 1.10 One-layer slab in a transient transfer – geometry

1.3.4 Case d: A One-layer Slab in Transient Transfer – the Analytical Quadrupole Matrix

A one-layer slab, made out of a homogeneous solid material, and shown in Figure 1.10, is characterized by its thickness e , its thermal conductivity λ and by its volumetric heat capacity ρc , ρ being its mass density and c its capacity. We call $a = \lambda/\rho c$ its thermal diffusivity. For a one-direction heat transfer the temperature field inside the slab can be written $T(x, t)$, where x is the coordinate perpendicular to the slab faces and t the time. We assume, furthermore, that the initial temperature in the slab is uniform and equal to T_0 . The temperature is the solution of the following heat equation:

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{a} \frac{\partial T}{\partial t} \quad \text{with} \quad T = T_0 \text{ for } t = 0 \quad (1.13)$$

Without loss of generality, it is possible to take an initial zero temperature ($T_0 = 0$). T is now the temperature variation with respect to the initial state. Since the problem is linear, a new temperature $T' = T - T_0$ with respect to equation (1.13), with a homogeneous initial condition, can always be defined.

The heat flux Φ at any location x inside the slab is defined for any cross-section S as

$$\Phi = -\lambda S \frac{\partial T}{\partial x} \quad (1.14)$$

The Laplace transforms of both temperature and flux are introduced:

$$\theta(x, p) = \int_0^\infty T(x, t) \exp(-pt) dt \quad \text{and} \quad \phi(x, p) = \int_0^\infty \Phi(x, t) \exp(-pt) dt \quad (1.15)$$

We now consider the Laplace transforms θ_1 and θ_2 of the front ($x = 0$) and rear ($x = e$) face temperatures, respectively, and the transforms ϕ_1 and ϕ_2 of the corresponding heat

fluxes. It can be shown – see Section 3.3.1 – that these four quantities are linked by the following equations:

$$\begin{bmatrix} \theta_1 \\ \phi_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} = \mathbf{M} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} \quad (1.16)$$

$$A = D = \cosh(ke); \quad B = \frac{1}{\lambda k S} \sinh(ke) \quad (1.17)$$

$$C = \lambda k S \sinh(ke), \quad k = \sqrt{p/a} \quad (1.18)$$

The coefficients A, B, C and D depend on the Laplace parameter p , on the thickness e of the slab and on the thermophysical properties of its constitutive material. It is interesting to note that equation (1.16) is valid for *any* boundary condition (in $x = 0$ and $x = e$). It is strictly equivalent to the heat equation and its associated boundary condition (1.13).

The matrix $\mathbf{M}$ with its four coefficients A, B, C and D completely characterizes the system here, in the same way as the thermal resistance R in steady-state heat transfer (case a) or as the heat capacity C_t for the lumped body assumption in transient transfer (case b). This characterization relates to an *external* representation of the system, since the four quantities $\theta_1, \theta_2, \phi_1$ and ϕ_2 are related to the external surfaces of the system and are not internal Laplace temperatures and fluxes. This type of representation is shown in Figure 1.11.

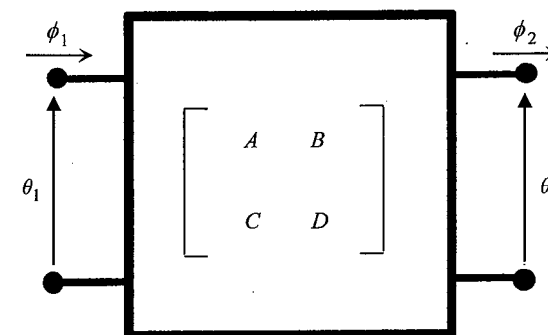


Figure 1.11 One-layer slab in a transient transfer – matrix representation

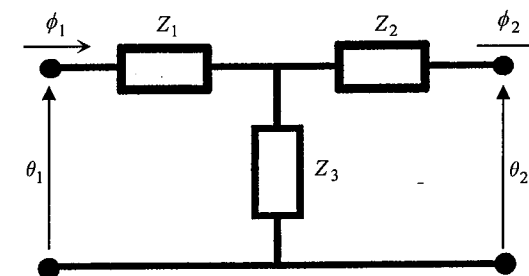


Figure 1.12 One-layer slab in a transient transfer – network representation

A network representation is also available (see Figure 1.12). It uses three thermal impedances Z_1, Z_2 and Z_3 that are related to the quadrupole coefficients A, B, C and D (see Section 3.3.1):

$$Z_1 = \frac{A-1}{C}; \quad Z_2 = \frac{D-1}{C}; \quad Z_3 = \frac{1}{C} \quad (1.19)$$

Up to now we have seen that, in a one-dimensional situation, the quadrupole matrix allows us to bridge the gap, in an exact way, between the resistance of the steady-state problem and the heat capacity of the lumped body assumption. The way it can be implemented will be now clarified using concrete examples.

1.4 HOW DOES IT WORK IN CONCRETE EXAMPLES?

1.4.1 Heat Pulse on an Insulated Slab

Equation (1.16) was valid for any type of boundary condition for a slab with parallel faces in a one-dimensional transient heat transfer. In order to calculate a temperature field, it is necessary to set two boundary conditions for this system.

It is now assumed that the front face of the slab ($x = 0$) is excited by a Dirac heat pulse of total energy density Q (in J m^{-2}) while its rear face ($x = e$) remains insulated. These boundary conditions are typical of the heat pulse method for the measurement of the thermal diffusivity a . In the time domain these conditions are

$$\Phi = QS\delta(t) \quad \text{at } x = 0 \quad (1.20)$$

$$\Phi = 0 \quad \text{at } x = e \quad (1.21)$$

where $\delta(t)$ is the Dirac distribution, which can physically be considered as the limit, as $\varepsilon \rightarrow 0$, of the rectangular function of width ε :

$$d_\varepsilon(t) = 0 \quad \text{for } t \leq 0 \text{ and for } t > \varepsilon \quad (1.22)$$

$$d_\varepsilon(t) = 1/\varepsilon \quad \text{for } 0 < t \leq \varepsilon \quad (1.23)$$

These boundary conditions can be translated into the Laplace domain (see Table 9.1 in Chapter 9 for the usual Laplace transforms):

$$\phi = QS \quad \text{at } x = 0 \quad (1.24)$$

$$\phi = 0 \quad \text{at } x = e \quad (1.25)$$

It should be noted that time t (in seconds) has been replaced by the Laplace parameter p (in hertz), but the space variable x remains unmodified.

The system (1.16) can be written as

$$\theta_1 = A\theta_2 + B\phi_2; \quad \phi_1 = C\theta_2 + D\phi_2 \quad (1.26)$$

where the Laplace boundary conditions (1.24) and (1.25), $\phi_1 = QS$ and $\phi_2 = 0$ have to be taken into account. Expression (1.26) therefore constitutes a linear system of two equations whose unknowns are the Laplace temperatures of the two faces, θ_1 and θ_2 . Its solution is

$$\theta_2 = \frac{QS}{C} = \frac{Q}{\lambda\sqrt{p/a} \sinh(e\sqrt{p/a})} \quad (1.27)$$

for the rear face Laplace temperature, and

$$\theta_1 = QS \frac{A}{C} = \frac{Q}{\lambda\sqrt{p/a} \tanh(e\sqrt{p/a})} \quad (1.28)$$

for the front face Laplace temperature.

Internal Laplace temperatures can also be found by the same technique. The preceding notations can be slightly modified at this point: the Laplace temperature and flux at any point x ($0 \leq x \leq e$) are now denoted by $\theta(x)$ and $\phi(x)$, respectively ($\theta(0) = \theta_1$, $\phi(0) = \phi_1$, $\theta(e) = \theta_2$, $\phi(e) = \phi_2$). If a layer of the slab defined by the coordinates x_1 and x_2 ($0 \leq x_1 < x_2 \leq e$) is considered, then we can write the following basic quadrupole equation:

$$\begin{bmatrix} \theta(x_1) \\ \phi(x_1) \end{bmatrix} = \begin{bmatrix} A(w) & B(w) \\ C(w) & D(w) \end{bmatrix} \begin{bmatrix} \theta(x_2) \\ \phi(x_2) \end{bmatrix} = \mathbf{M} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} \quad (1.29)$$

where the four coefficients A, B, C and D are defined by equations (1.17) and (1.18), simply by replacing the thickness of the slab e by the thickness of the layer w (with $w = x_2 - x_1$).

Writing this equation for $x_1 = e/2$ and $x_2 = e$, it is possible to calculate the midslab Laplace temperature and flux:

$$\theta(e/2) = A(e/2)\theta(e) = \frac{Q}{2\lambda\sqrt{p/a} \sinh\left(\frac{e}{2}\sqrt{p/a}\right)} \quad (1.30)$$

$$\phi(e/2) = C(e/2)\theta(e) = \frac{QS}{2} \frac{1}{\cosh\left(\frac{e}{2}\sqrt{p/a}\right)} \quad (1.31)$$

The preceding three temperatures and the midslab flux can be calculated, in the time domain, by numerical inversion of the Laplace transforms (1.27), (1.28), (1.30) and (1.31).

Numerical inversions of these transforms are presented in detail in Chapter 9. We have decided to use Stehfest's algorithm here (see Section 9.3.2). If $F(p)$ is the known Laplace transform of the function $f(t)$, then an approximate value of this function at time t can be calculated:

$$f(t) \cong \frac{\ln(2)}{t} \sum_{i=1}^n \nu_i F(i \ln(2)/t) \quad (1.32)$$

The coefficients ν_i are given in Chapter 9 for different values of the total number of terms n .

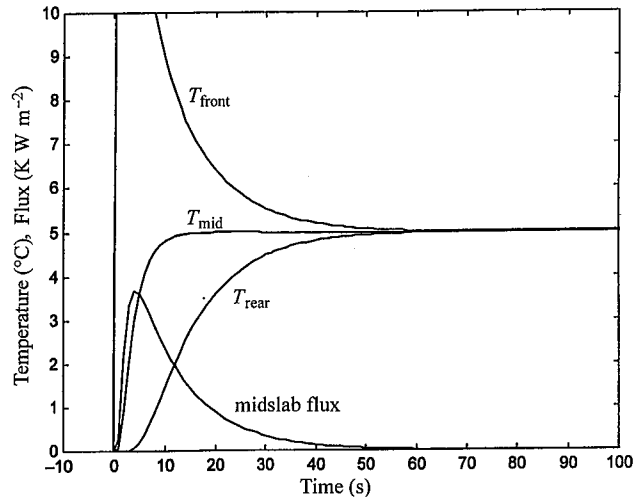


Figure 1.13 Temperature and responses of a one-layer slab excited by a front face Dirac heat flux

The calculated variations with time of the rear and front face temperatures and of the midslab temperature and flux (with $n = 10$), are plotted in Figure 1.13. They correspond to the following values of the parameters:

$$e = 0.01 \text{ m}; \quad \lambda = 2 \text{ W m}^{-1} \text{ K}^{-1}; \quad a = 10^{-6} \text{ m}^2 \text{ s}^{-1}; \quad Q = 10^5 \text{ J m}^{-2}; \quad S = 1 \text{ m}^2$$

The corresponding code, in MATLAB[®] language, is given in Appendix 1.1.

Remark 1. An exact analytical solution of the front and rear face temperatures is available here:

$$T_2(t) = \frac{Q}{\rho c e} \left[1 + 2 \sum_{k=1}^{\infty} (-1)^k \exp(-k^2 \pi^2 a t / e^2) \right] \quad (1.33)$$

$$T_1(t) = \frac{Q}{\rho c e} \left[1 + 2 \sum_{k=1}^{\infty} \exp(-k^2 \pi^2 a t / e^2) \right] \quad (1.34)$$

These two equations correspond to local values of the general case exact solution which can be calculated by the method of separation of the variables:

$$T(x, t) = \frac{Q}{\rho c e} \left[1 + 2 \sum_{k=1}^{\infty} \cos\left(k \pi \frac{x}{e}\right) \exp(-k^2 \pi^2 a t / e^2) \right] \quad (1.35)$$

1.4.2 Analytical Asymptotic Solutions

The analytical explicit expression for a temperature or flux in the Laplace domain can introduce an interesting expression for limit cases, i.e. cases where the variable, time here, is

small or large or for cases where one parameter formally present in this expression takes a particular (large of small) value.

Short time solutions

It is possible, for example, to get an approximate form of the front face temperature $T_1(t)$ that is valid for small time values (an asymptotic solution). If t tends to zero in the time domain, then one must consider the limit of $\theta_1(p)$, given by equation (1.28), as p goes to infinity (this property of the Laplace transform is given by equation (9.24) in Chapter 9).

Using the fact that the hyperbolic tangent present in equation (1.28) goes to unity, this equation becomes

$$\theta_1(p) \cong \frac{Q}{b \sqrt{p}} \quad \text{as } p \rightarrow \infty \quad (1.36)$$

where $b = (\lambda \rho c)^{1/2}$ is the thermal effusivity of the material that constitutes the slab.

The inverse Laplace transform of equation (1.36) is known explicitly (see Table 9.1):

$$T_1(t) \cong \frac{Q}{b \sqrt{\pi t}} \quad \text{as } t \rightarrow 0 \quad (1.37)$$

It should be noted at this point that neither the analytical exact solution in the time domain (1.34), nor any numerical solution of the heat equation, could have given this short time approximation.

Remark 2. One might consider that equation (1.37) corresponds also to the front face temperature response of a semi-infinite medium (i.e. a medium whose extent is only limited by one face, here at $x = 0$, while the other face is pushed away to infinity). This can be shown by making the thickness e of the slab go to infinity in equation (1.28), which mechanically yields the same expression (1.36) for $\theta_1(p)$.

A short time approximate solution can also be found for the rear face temperature starting from equation (1.27). Thus, for large p -values, the negative exponential in the hyperbolic sine definition can be ignored with respect to the corresponding positive exponential, i.e.

$$\theta_2(p) \cong \frac{2Q}{\lambda \sqrt{p/a} \exp(e \sqrt{p/a})} \quad \text{as } p \rightarrow \infty \quad (1.38)$$

At this point it is necessary to define a reduced Laplace parameter $p^* = e^2 p/a$ that corresponds to the reduced time, the Fourier number $t^* = at/e^2$. The rear face Laplace temperature can then be written as

$$\theta_2(p) = L[T_2(t)] = \int_0^{\infty} T_2(t) \exp(-pt) dt = \frac{e^2}{a} \int_0^{\infty} T_2 \exp(-p^* t^*) dt^* = \frac{e^2}{a} L^*[T_2] \quad (1.39)$$

Consequently, a modified Laplace temperature $\theta_2^* = a\theta_2/e^2$ can now be defined, which is the transform of the temperature that is now considered as a function of the Fourier number. We obtain:

$$\theta_2^*(p^*) \cong \frac{2aQe}{e^2\lambda\sqrt{p^*}} \exp(-\sqrt{p^*}) = \frac{2Q}{\rho ce} \frac{\exp(-\sqrt{p^*})}{\sqrt{p^*}} \quad \text{as } p^* \rightarrow \infty \quad (1.40)$$

The inverse Laplace transform of this function is given in Table 9.1 in Chapter 9:

$$T_2 \cong \frac{2Q}{\rho ce} \frac{1}{\sqrt{\pi t^*}} \exp\left(-\frac{1}{4t^*}\right) \quad \text{as } t^* \rightarrow 0 \quad (1.41)$$

Long time solutions

The same kind of procedure as shown above can be adopted to derive rear or front face temperatures for long time values. In this case the Laplace parameter p has to go to zero. Since both the 'sinh' and 'tanh' functions can be approximated by the value of their argument, $e\sqrt{p/a}$ in this case, equations (1.27) and (1.28) produce the same approximation:

$$\theta_2(p) \cong \theta_1(p) \cong \frac{Q}{\rho cep} \quad \text{as } p \rightarrow 0 \quad (1.42)$$

The Laplace inversion is straightforward:

$$T_2(t) \cong T_1(t) \cong \frac{Q}{\rho ce} = T_\ell \quad \text{as } t \rightarrow \infty \quad (1.43)$$

This common limit for both the rear and the front face temperatures deserves a physical interpretation: since the slab is insulated from the outside (except of course from the heat pulse excitation that constitutes the front face surface source – see Chapter 2) the total heat QS , coming from the heat pulse excitation, corresponds to an increase in enthalpy $\rho ceS T_\ell$ of the material (first principle). For long time values the temperature becomes uniform in the slab (which is insulated, except from the flash), at this T_ℓ level, and thus no longer can heat be transferred by conduction (no temperature gradient).

1.4.3 Other Types of Transient Excitations

In the above section we considered $T_1(t)$ or $T_2(t)$ as the response to a front face excitation $\Phi_1(t)$ (for an insulated rear face: $\Phi_2(t) = 0$). This can be represented in terms of an input/output linear system. For example, if $T_1(t)$ is the response of the system, then we can write, according to equation (1.26):

$$\theta_1 = \frac{A}{C} \phi_1 = L[f(t)]L[\Phi_1(t)] \quad (1.44)$$

where $f(t)$ is the inverse Laplace transform of A/C .

The simple product (1.44) in the Laplace domain becomes a convolution product in the time domain:

$$T_1(t) = \int_0^t f(t-\tau)\Phi_1(\tau)d\tau \quad (1.45)$$

$f(t)$ is the impulse response of the system (here the homogeneous slab). It is given by equation (1.34), while A/C is given by equations (1.17) and (1.18) if we consider a pulse of unit energy ($Q = 1 \text{ J m}^{-2}$). $f(t)$ is the Green's function corresponding to this geometry and to these boundary conditions (insulated rear face, flux excitation on the front face and temperature response of the front face). It can be considered as the first brick in the construction of the response to any kind of excitation $\Phi_1(t)$. The interested reader can consult Beck *et al.* (1992) for this type of method for constructing analytical solutions to heat conduction problems. Equation (1.44) shows a very interesting feature: it replaces the convolution product (1.45) in the time domain by a simple product in the Laplace domain. Responses to other types of excitations can be considered. For example, starting from thermal equilibrium at time $t = 0$, the same slab can be stimulated by a sudden constant heat flux excitation starting at this time, i.e.

$$\begin{cases} \Phi_1(t) = 0 & \text{for } t \leq 0 \\ \Phi_1(t) = WS & \text{for } t > 0 \end{cases} \quad (1.46)$$

where W is the power per unit surface of the excitation. Since the transform of $\Phi_1(p)$ is simply W/p , the front face response is

$$\theta_1 = \frac{WS}{p} \frac{A}{C} \phi_1 = \frac{W}{\lambda p \sqrt{p/a} \tanh(e\sqrt{p/a})} \quad (1.47)$$

Long and short times solutions can be easily obtained:

$$\theta_1 \underset{\text{as } p \rightarrow 0}{\cong} \frac{W}{\lambda p \sqrt{p/a} e \sqrt{p/a}} = \frac{W}{\rho cep^2} \Rightarrow T_1(t) \underset{\text{as } t \rightarrow \infty}{\cong} \frac{Wt}{\rho ce} \quad (1.48)$$

$$\theta_1 \underset{\text{as } p \rightarrow \infty}{\cong} \frac{W}{\lambda p \sqrt{p/a}} = \frac{W}{bp^{3/2}} \Rightarrow T_1(t) \underset{\text{as } t \rightarrow 0}{\cong} \frac{2W}{b\sqrt{\pi}} \sqrt{t} \quad (1.49)$$

These approximate asymptotic behaviours show that the front face response begins to be proportional to the square root of time, while it exhibits a variation nearly proportional to time as constant heat flux heating goes on.

The calculated variations with time of the front face temperature (Laplace Stehfest's inversion of equation (1.47) with $n = 10$) as well as its two asymptotic behaviours (1.48) and (1.49) are plotted in Figure 1.14. They correspond to the following values of the parameters:

$$e = 0.01 \text{ m}; \quad \lambda = 2 \text{ W m}^{-1} \text{ K}^{-1}; \quad a = 10^{-6} \text{ m}^2 \text{ s}^{-1}; \quad W = 10^4 \text{ W m}^{-2}$$

The corresponding code, in MATLAB[®] language, is given in Appendix 1.2.

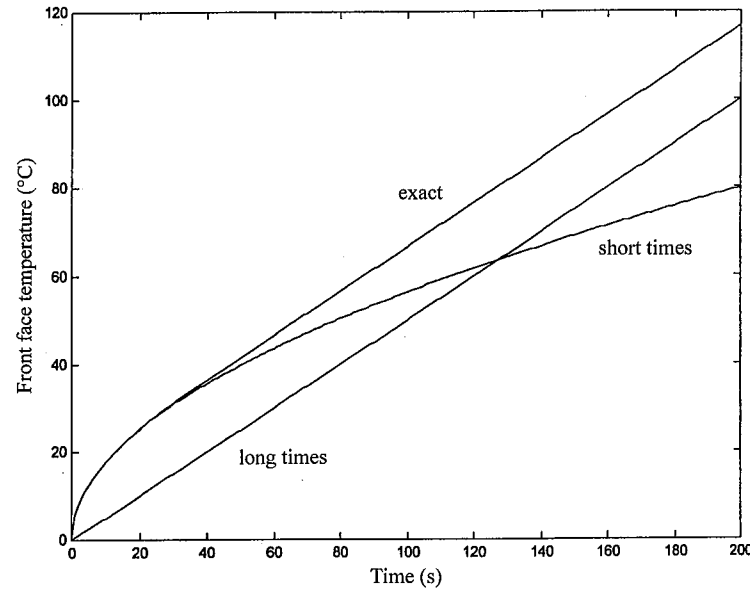


Figure 1.14 Front face temperature response of a semi-infinite medium excited by a front face step heat flux

1.4.4 Quadrupole Toolbox for Multimaterials

In the preceding sections it was shown that the quadrupole method can provide analytical explicit solutions, in the Laplace domain, for classical cases of transient conduction heat transfer through a one-layer slab. One major feature of this method is the possibility of producing such a solution for multimaterials, i.e. materials that are made out of more than one material.

We will illustrate this point through two examples that concern stratified media. They will show that each homogeneous layer, boundary condition (heat exchange coefficient), interface condition (thermal contact resistance) and macroscopic constriction (change of section, if the average surface temperatures are considered) can be represented by a quadrupole matrix. The product matrix, i.e. the product of the preceding matrices in cascade, links the Laplace temperature vectors of the two sides of this stratified medium. It can also be represented by an equivalent electrical circuit using the notion of thermal impedance (in the Laplace domain).

Two-layer building wall in a transient regime

A wall, of cross-section area S , separates the inside of a building from the outside. It is composed of two homogeneous solid media characterized here by subscripts 1 and 2 – see Figure 1.15. Let e_j be the thickness of each layer ($j = 1$ or 2) and λ_j , ρc_j and $a_j = \lambda_j / \rho c_j$ be their thermal conductivities, volumetric heat and thermal diffusivities, respectively. One-directional heat transfer in the x -direction is considered here. Transfers by convection on the

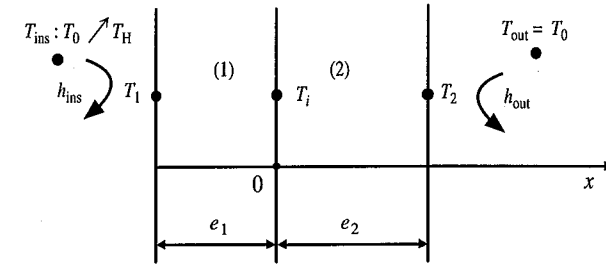


Figure 1.15 Two-layer wall in a transient regime

inside and outside faces of the wall are characterized by two transfer coefficients h_{ins} and h_{out} . The inside and outside air temperatures of the building are denoted T_{ins} and T_{out} . Before time $t = 0$, both the wall and environment (inside and outside) are isothermal at temperature T_0 . At time $t = 0$, the inside temperature is instantaneously raised to a level T_H , while the outside temperature remains constant at level T_0 . The problem is to find the temperature variation with time of the internal face, $T_1(t)$, of the external face, $T_2(t)$, and of the interface between the two solid layers, $T_i(t)$. We consider here the Laplace transforms θ of the translated temperature $\theta = L[T - T_0]$. The Laplace temperature-vector corresponding to the inside air temperature and to the flux penetrating the internal face of the wall is linked to the same vector composed of the outside air temperature and to the flux leaving the external face by the following matrix equation:

$$\begin{bmatrix} \theta_{ins} \\ \phi_{ins} \end{bmatrix} = \mathbf{H}_{ins} \mathbf{M} \mathbf{H}_{out} \begin{bmatrix} \theta_{out} \\ \phi_{out} \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \theta_{out} \\ \phi_{out} \end{bmatrix} \quad (1.50)$$

with

$$\mathbf{M} = \mathbf{M}_1 \mathbf{M}_2 = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

where the different matrices can be defined as

$$\mathbf{M}_1 = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix}; \quad \mathbf{M}_2 = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix};$$

$$\mathbf{H}_{ins} = \begin{bmatrix} 1 & 1/(h_{ins} S) \\ 0 & 1 \end{bmatrix}; \quad \mathbf{H}_{out} = \begin{bmatrix} 1 & 1/(h_{out} S) \\ 0 & 1 \end{bmatrix} \quad (1.51)$$

with

$$\left. \begin{aligned} A_j &= D_j = \cosh(e_j \sqrt{p/a_j}); & B_j &= \frac{1}{\lambda_j S \sqrt{p/a_j}} \sinh(e_j \sqrt{p/a_j}) \\ C_j &= \lambda_j S \sqrt{p/a_j} \sinh(e_j \sqrt{p/a_j}) & \text{with } a_j &= \frac{\lambda_j}{\rho c_j} \end{aligned} \right\} \text{ for } j = 1 \text{ or } 2 \quad (1.52)$$

The matrices M_1 and M_2 correspond to the two layers (1) and (2), respectively. They are of the homogeneous slab type, already given by equations (1.16)–(1.18). The matrices H_{ins} and H_{out} are matrices corresponding to the inside and outside heat transfer coefficients respectively. They correspond to resistive quadrupoles of the form already given by equations (1.1)–(1.3).

Equation (1.50) constitutes a system of two equations with two unknowns, ϕ_1 and ϕ_2 , where the known data are the transforms of the translated inside and outside air temperatures: $\theta_{ins} = (T_H - T_0)/p$ and $\theta_{out} = 0$. These Laplace fluxes are therefore equal to

$$\phi_{out} = \frac{1}{\mathcal{B}} \theta_{ins} = \frac{1}{\mathcal{B}} \frac{T_H - T_0}{p} \quad (1.53)$$

$$\phi_{ins} = \mathcal{D} \phi_{out} = \frac{\mathcal{D}}{\mathcal{B}} \frac{T_H - T_0}{p} \quad (1.54)$$

where the coefficients $\mathcal{B}$ and $\mathcal{D}$ can be calculated starting from the coefficients of the two-layer wall matrix M – see equation (1.50):

$$\mathcal{B} = \frac{A}{h_{out} S} + B + \frac{C}{h_{out} h_{ins} S^2} + \frac{D}{h_{ins} S}; \quad \mathcal{D} = \frac{C}{h_{out} S} + D \quad (1.55)$$

with

$$\begin{aligned} A &= A_1 A_2 + B_1 C_2; & B &= A_1 B_2 + B_1 A_2 \\ C &= C_1 A_2 + A_1 C_2; & D &= C_1 B_2 + A_1 A_2 \end{aligned} \quad (1.56)$$

The four poles of equation (1.50) are now known, and it is therefore possible to calculate the Laplace temperature flux vectors at the internal face (1) of the wall, at the interface (i) of the two layers, and at the external face (2) of the wall:

$$\begin{bmatrix} \theta_1 \\ \phi_1 \end{bmatrix} = M H_{out} \begin{bmatrix} \theta_{out} \\ \phi_{out} \end{bmatrix}; \quad \begin{bmatrix} \theta_i \\ \phi_i \end{bmatrix} = M_2 H_{out} \begin{bmatrix} \theta_{out} \\ \phi_{out} \end{bmatrix}; \quad \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} = H_{out} \begin{bmatrix} \theta_{out} \\ \phi_{out} \end{bmatrix} \quad (1.57)$$

These three Laplace temperatures are

$$\theta_1 = \left(\frac{A}{h_{out} S} + B \right) \phi_{out}; \quad \theta_i = \left(\frac{A_2}{h_{out} S} + B_2 \right) \phi_{out}; \quad \theta_2 = \frac{1}{h_{out} S} \phi_{out} \quad (1.58)$$

The calculated variations with time of the temperatures of the internal face $T_1(t)$, of the interface $T_i(t)$ and of the outside face $T_2(t)$ can be calculated using Stehfest's method of inversion of equations (1.58). They are plotted in Figure 1.16. They correspond to the following values of the parameters:

$$\begin{aligned} e_1 &= 0.20 \text{ m}; & \lambda_1 &= 1.7 \text{ W m}^{-1} \text{ K}^{-1}; & a_1 &= 1.2 \cdot 10^{-6} \text{ m}^2 \text{ s}^{-1} \\ e_2 &= 0.05 \text{ m}; & \lambda_2 &= 0.16 \text{ W m}^{-1} \text{ K}^{-1}; & a_2 &= 1.6 \cdot 10^{-7} \text{ m}^2 \text{ s}^{-1} \\ T_H - T_0 &= 15^\circ \text{C}; & h_{ins} &= 10 \text{ W m}^{-2} \text{ K}^{-1}; & h_{out} &= 10 \text{ W m}^{-2} \text{ K}^{-1} \end{aligned}$$

The corresponding code, in MATLAB[®] language, is given in Appendix 1.3

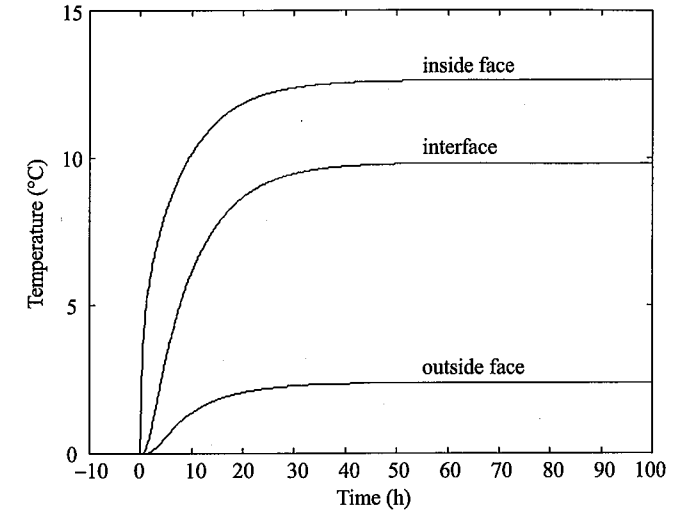


Figure 1.16 Two-layer slab in a transient regime: temperature variations

Remark 3. The possibility of calculating the Laplace temperature in this two-layer wall is not limited to the particular points located at the faces of the two layers. If we consider any point x , located for example inside layer (1) ($-e_1 \leq x \leq 0$), then its Laplace temperature can be obtained in a very straightforward way:

$$\begin{bmatrix} \theta(x) \\ \phi(x) \end{bmatrix} = M_1(-x) M_2 H_{out} \begin{bmatrix} 0 \\ \phi_{out} \end{bmatrix} \quad (1.59)$$

where the matrix $M_1(-x)$ corresponds to the matrix M_1 , where the thickness e_1 has been replaced by $(-x)$ in its coefficients given by equation (1.52).

Electronic chip cooling

The complete assembly is shown in Figure 1.17. An electronic chip generates heat at the top of the device: a time-constant heat flux density heating φ_0 (W m^{-2}), uniform on the cross-section S , starts at time $t = 0$. The corresponding flux flows through two layers: a ceramic layer (thickness e_{ce} , conductivity λ_{ce} and volumetric heat capacity ρc_{ce}) and an aluminium layer (thickness e_{al} , conductivity λ_{al} and volumetric heat capacity ρc_{al}). It is dissipated in ambient air, at constant temperature T_∞ , by the means of an aluminium fin of infinite extent (constant cross-section s and perimeter m). There is an imperfect thermal contact between the ceramic layer and the aluminium layer that can be characterized by a thermal contact resistance R_c (in K W^{-1}).

The convective heat transfer coefficient h is assumed to be uniform on the lateral surface of the fin and both the top and bottom surfaces (except the fin) are assumed to be insulated. The whole system is at thermal equilibrium with the ambient air at T_∞ before heating starts. We consider the Laplace transforms θ of the translated temperature $\theta = L[T - T_\infty]$. The whole system constitutes a flux pipe, from the heated top surface (denoted by subscript 1) to

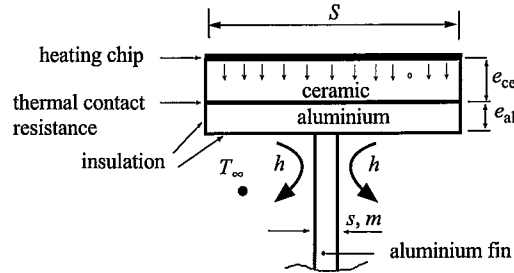


Figure 1.17 Cooling device of an electric chip

the lateral surface of the fin and the surrounding air (denoted by subscript 2). In the Laplace domain, two components of the temperature flux vectors are known, while the other components are unknown (θ_1 and ϕ_2). The Laplace transform θ_1 of the front face temperature is of prime practical interest since it allows the calculation of the temperature level $T_1(t)$ of the chip. The following matrix relationship between the Laplace temperature-flux vectors can be written:

$$\begin{bmatrix} \theta_1 \\ \phi_1 \end{bmatrix} = M_{ce} R M_{al} M_{constr} M_{fin} \begin{bmatrix} \theta_2 \\ \phi_2 \end{bmatrix} \quad (1.60)$$

where the different matrices can be defined:

$$\begin{aligned} M_{ce} &= \begin{bmatrix} A_{ce} & B_{ce} \\ C_{ce} & D_{ce} \end{bmatrix}; & R &= \begin{bmatrix} 1 & R_c \\ 0 & 1 \end{bmatrix}; & M_{al} &= \begin{bmatrix} A_{al} & B_{al} \\ C_{al} & D_{al} \end{bmatrix} \\ M_{constr} &= \begin{bmatrix} 1 & Z_{constr} \\ 0 & 1 \end{bmatrix}; & M_{fin} &= \begin{bmatrix} 1 & Z_{fin} \\ 0 & 1 \end{bmatrix} \end{aligned} \quad (1.61)$$

with

$$\left. \begin{aligned} A_\alpha &= D_\alpha = \cosh(e_\alpha \sqrt{p/a_\alpha}); & B_\alpha &= \frac{1}{\lambda_\alpha S \sqrt{p/a_\alpha}} \sinh(e_\alpha \sqrt{p/a_\alpha}) \\ C_\alpha &= \lambda_\alpha S \sqrt{p/a_\alpha} \sinh(e_\alpha \sqrt{p/a_\alpha}) & \text{with } a_\alpha &= \frac{\lambda_\alpha}{\rho c_\alpha} \end{aligned} \right\} \text{ for } \alpha = \text{ce or al} \quad (1.62)$$

and

$$Z_{constr} = \frac{8}{\pi \lambda_{al} (3\sqrt{\pi s} + 8s\sqrt{p/a_{al}})}; \quad Z_{fin} = \frac{1}{\lambda_{al} s \sqrt{\frac{p}{a_{al}} + \frac{hm}{\lambda_{al} s}}} \quad (1.63)$$

The matrices M_{ce} and M_{al} correspond to the ceramic and aluminium layers, respectively. They are both quadrupole matrices of a homogeneous plane slab whose expression has already been given by equations (1.16)–(1.18). Since aluminium has a very high

conductivity, the aluminium layer can also be represented by the approximation of a 'capacitive' matrix (lumped body approximation) corresponding to the form already given by equation (1.8):

$$M_{al} = \begin{bmatrix} A_{al} & B_{al} \\ C_{al} & D_{al} \end{bmatrix} \cong \begin{bmatrix} 1 & 0 \\ C_{tal} p & 1 \end{bmatrix} \quad \text{with } C_{tal} = \rho c_{al} e_{al} S \quad (1.64)$$

The matrix M_{constr} takes into account the two- or three-dimensional constriction of the flux lines in the aluminium layer: it corresponds to a macroscopic narrowing of the flux pipe where the heat flux lines have to pass from a wide cross-section (S) to a narrow cross-section (s). The expression for the coefficient Z_{constr} , the constriction impedance, is derived in Section 6.1 (see equation (6.63)). It is valid for $s/S \ll 1$ and $\sqrt{s}/e_{al} \ll 1$.

The matrix M_{fin} models the heat flow through the fin in a transient transfer. The expression for the coefficient Z_{fin} , the impedance of the fin, is derived in Section 3.4.2 (see equation (3.150)).

The matrix R represents, through its coefficient R_c , the contact resistance between the ceramic and aluminium layers. This resistance takes into account the imperfect microscopic contact of this interface (microconstrictions and heat flow through microasperities). This matrix is a 'resistive' quadrupole matrix that has the structure given by equation (1.3).

Remark 4. Because of the presence of the reduction in the cross-section at the level of the aluminium layer–aluminium fin interface, the heat transfer is no longer one-dimensional in this example. We must therefore consider all the Laplace temperatures that appear in the different quadrupole equations, such as in equation (1.60), as transforms of the average temperature over the cross-section (S for T_1 and s for T_2). This explains the use of the constriction impedance.

The product matrix can then be defined as

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = M_{ce} R M_{al} M_{constr} M_{fin} \quad (1.65)$$

Before taking into account the boundary conditions, it is possible to represent this system by a four-terminal network of impedances, as shown in Figure 1.18. The three impedances associated with the ceramic layer depend on the coefficients of its matrix according to equation (1.19):

$$Z_{1ce} = Z_{2ce} = \frac{A_{ce} - 1}{C_{ce}}; \quad Z_{3ce} = \frac{1}{C_{ce}} \quad (1.66)$$

With the particular boundary conditions that are taken here ($\phi_1 = \varphi_0 S/p$, $\theta_2 = 0$), it is easy to calculate the transform of the chip's temperature simply by applying equation (1.60):

$$\theta_1 = \frac{B}{D} \phi_1 = \frac{(Z_{constr} + Z_{fin})(A_{ce} + p C_{tal}(A_{ce} R_c + B_{ce})) + A_{ce} R_c + B_{ce}}{(Z_{constr} + Z_{fin})(C_{ce} + p C_{tal}(C_{ce} R_c + A_{ce})) + C_{ce} R_c + A_{ce}} \frac{\varphi_0 S}{p} \quad (1.67)$$

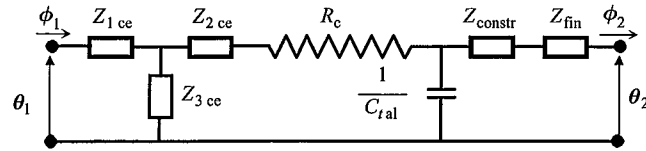


Figure 1.18 Four-terminal impedance network of the electronic chip without any boundary conditions

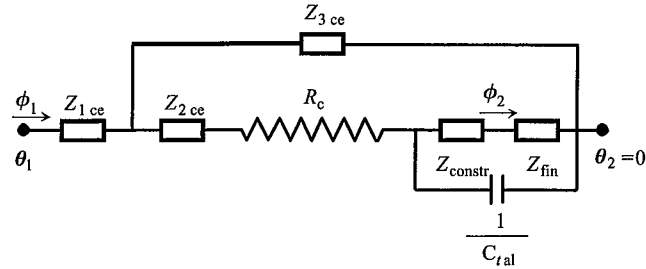


Figure 1.19 Two-terminal network of the electronic chip (for constant air temperature)

It is interesting to note at this point that the B/D ratio represents the transfer function of the system, where the chip's temperature is the output, for any kind of power input, when the temperature of the air stays constant and equal to T_∞ (and consequently $\theta_2 = 0$). It is possible to take into account this condition in the four-terminal network of impedances shown in Figure 1.18: since the potential $\theta_2 = 0$, it is possible to connect the two terminals (the black full circles) on the right of this figure (as in an electric circuit) to show that the upper right terminal is at the same zero potential as the zero potential of the lower base line (this is the thermal analogue to an electrical bypass). The resulting diagram is the two-terminal network represented in Figure 1.19. We consider, in an analogous representation, that the flux ϕ_1 flows through a temperature difference θ_1 , the corresponding impedance B/D being calculated through the classical expressions that allows us to calculate equivalent impedances of circuits in parallel or in series:

$$\frac{B}{D} = Z_{1ce} + \left(\frac{1}{Z_{3ce}} + \frac{1}{Z_{2ce} + R_c + \left(\frac{1}{Z_{constr} + Z_{fin}} + pC_{tal} \right)^{-1}} \right)^{-1} \quad (1.68)$$

Of course, equation (1.68) gives exactly the same value for the impedance B/D as equation (1.67).

The steady-state regime, which is reached for long times, can be calculated by the limit of equation (1.67) – or (1.68) – as the Laplace parameter p goes to zero:

$$\begin{aligned} A_{ce} &\rightarrow 1; & B_{ce} &\rightarrow \frac{e_{ce}}{\lambda_{ce} S} = R_{ce}; & C_{ce} &\rightarrow 1; & pC_{tal} &\rightarrow 0 \\ Z_{constr} &\rightarrow \frac{8}{3\pi^{3/2}\lambda_{al}\sqrt{s}} = R_{constr}; & Z_{fin} &\rightarrow \frac{1}{\sqrt{\lambda_{al} h m s}} = R_{fin} \end{aligned}$$

where R_{ce} , R_{constr} and R_{fin} are the thermal resistances of the ceramic layer, the constriction and the fin, respectively. Equation (1.67) then becomes:

$$\theta_1 \underset{\text{as } p \rightarrow 0}{\cong} (R_{constr} + R_{fin} + R_c + R_{ce}) \frac{\varphi_0 S}{p} \quad \text{and} \quad \theta_1 \underset{\text{as } p \rightarrow 0}{\cong} \frac{T_1 - T_\infty}{p} \quad (1.69)$$

which gives the long time temperature difference between the chip and ambient air:

$$T_1 - T_\infty = R\varphi_0 S \quad \text{with } R = R_{ce} + R_c + R_{constr} + R_{fin} \quad (1.70)$$

This last result could have been derived directly from Figure 1.19: as time goes to infinity (i.e. as p goes to zero), the capacity impedance $1/pC_{tal}$ goes to infinity as well as the Z_{3ce} impedance. The fluxes passing through the two corresponding branches of the circuit go to zero, which means that these branches can be 'cut'. We find therefore a unique branch with impedances Z_{1ce} and Z_{2ce} (which both tend to $R_{ce}/2$) and resistances R_c , R_{constr} and R_{fin} in series. This leads directly to the equivalent resistance of the steady-state circuit already given by equation (1.70).

1.5 INTEREST OF ANALYTICAL METHODS

The quadrupole method yields explicit solutions in the Laplace domain. This analytical property can be very valuable if one tries to construct models for estimating thermal properties starting from one experiment. We present two examples that explain this point here.

1.5.1 The Heat Pulse Experiment with Losses

In a flash experiment to measure thermal diffusivity, the measured signal is the temperature of the rear face (subscript e) of a cylindrical sample (of thickness e , diffusivity a and conductivity λ) which is submitted to a heat pulse of total absorbed energy Q (in J m^{-2}) on its front face (subscript 0). The method for obtaining the Laplace temperature of the rear face is developed in Problem 3.2 in Section 3.6. The reduced Laplace temperature of the rear face is

$$\theta_e^* = \frac{1}{C^* + A^*(H_0 + H_e) + B^*H_0H_e} \quad (1.71)$$

where θ_e^* is a function of the reduced Laplace parameter $p^* = ap/e^2$, A^* , B^* and C^* being the coefficients of the reduced matrix of the slab's quadrupole:

$$A^* = \cosh(\sqrt{p^*}); \quad B^* = \frac{1}{\sqrt{p^*}} \sinh(\sqrt{p^*}); \quad C^* = \sqrt{p^*} \sinh(\sqrt{p^*}) \quad (1.72)$$

The Biot numbers $H_0 = h_0e/\lambda$ and $H_e = h_ee/\lambda$ represent the reduced heat transfer coefficients of the losses h_0 and h_e (convection and linearized radiation) on the front and rear

faces, respectively. Equation (1.71) clearly shows that the reduced temperature formally depends on three parameters: the characteristic time e^2/a , and the two Biot numbers H_0 and H_e . This equation shows that both heat loss coefficients have a symmetrical effect on the measured signal. This result is not obvious given the non-symmetrical aspect of the thermal excitation that is applied to the front face only. This property could not have been shown by a rear face temperature solution produced by the method of separation of variables. For moderate values of the ambient temperature, both Biot numbers H_0 and H_e are lower than 0.1 and the last term at the denominator can be ignored (for large p^* -values, i.e. for low reduced times or Fourier numbers $t^* = e^2 t/a$). This means that there is no point in considering two different Biot number in the parameter estimation problem starting from an experimental signal, since the signal is sensitive to their sum only.

1.5.2 Non-Destructive Thermal Testing of Delaminations in Stratified Media

A slab of thickness e is made out of a stratified composite medium (superposition of plies of carbon fibres that are held by a polymerized resin matrix). From a macroscopic point of view the slab can be considered as a homogeneous anisotropic material, of transverse conductivity λ_x and of in-plane longitudinal conductivity λ_r .

The delamination in such a slab, i.e. a partial unsticking between two plies, can be characterized thermally by a thermal contact resistance R_c located at some depth e_1 inside the material. If the front face is excited by a heat pulse of energy density Q (J m^{-2}) at time $t = 0$, then a one-dimensional transfer will take place inside the slab – see Figure 1.20.

We further assume that the faces are insulated and that the initial temperature in the slab is uniform (and equal to zero). The temperatures T_f and T_r of the front face ($x = 0$) and of the rear face ($x = e_1 + e_2 = e$) can be measured. Their Laplace transforms θ_f and θ_r can be calculated if we consider the quadrupole whose matrix is the product of the homogeneous material first layer matrix (subscript 1), of the resistive quadrupole matrix produced by the thermal contact resistance R_c , and of the homogeneous material second layer matrix (subscript 2):

$$\begin{bmatrix} \theta_f \\ Q \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & A_1 \end{bmatrix} \begin{bmatrix} 1 & R_c \\ 0 & 1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & A_2 \end{bmatrix} \begin{bmatrix} \theta_r \\ 0 \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \theta_r \\ 0 \end{bmatrix} \quad (1.73)$$

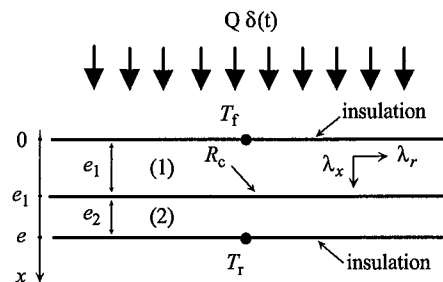


Figure 1.20 Non-destructive thermal evaluation of a delamination

with

$$\left. \begin{aligned} A_j &= \cosh(e_j \sqrt{p/a_x}); & B_j &= \frac{1}{\lambda_x \sqrt{p/a_x}} \sinh(e_j \sqrt{p/a_x}) \\ C_j &= \lambda_x \sqrt{p/a_x} \sinh(e_j \sqrt{p/a_x}) \end{aligned} \right\} \text{ for } j = 1 \text{ or } 2$$

Only the transverse diffusivity $a_x = \lambda_x / \rho c$, where ρc is the volumetric heat capacity of the material, is active here since the heat transfer is one-dimensional. We have considered a unit slab of area $S = 1 \text{ m}^2$, which implies that the fluxes are in fact heat flux densities and that the thermal contact resistance R_c is expressed in $\text{K m}^2 \text{ W}^{-1}$.

The Laplace temperatures of the two faces are

$$\theta_f = \frac{\mathcal{A}}{\mathcal{C}} Q = \frac{A + A_1 C_2 R_c}{C + C_1 C_2 R_c} Q; \quad \theta_r = \frac{1}{\mathcal{C}} Q = \frac{1}{C + C_1 C_2 R_c} Q \quad (1.74)$$

where $A = A_1 A_2 + B_1 C_2 = \cosh(e \sqrt{p/a_x})$ and $C = C_1 A_2 + A_1 C_2 = \lambda_x \sqrt{p/a_x} \sinh(e \sqrt{p/a_x})$ are two coefficients of the sound slab quadrupole matrix.

The same problem can be solved for a sound slab ($R_c = 0$, subscript 0):

$$\theta_{f0} = \frac{A}{C} Q; \quad \theta_{r0} = \frac{1}{C} Q \quad (1.75)$$

This allows us to define a front and a rear face thermal contrast, i.e. the difference between the temperatures on a defective slab and on a sane slab. The transforms of these two contrasts are

$$\begin{aligned} \Delta\theta_f &= L[T_f(t) - T_{f0}(t)] = \theta_f - \theta_{f0} \\ &= \frac{QR_c \sinh^2(ke_2)}{\sinh(ke)(\sinh(ke) + \lambda_x k R_c \sinh(ke_1) \sinh(ke_2))} \end{aligned} \quad (1.76)$$

and

$$\begin{aligned} \Delta\theta_r &= L[T_r(t) - T_{r0}(t)] = \theta_r - \theta_{r0} \\ &= \frac{QR_c \sinh(ke_1) \sinh(ke_2)}{\sinh(ke)(\sinh(ke) + \lambda_x k R_c \sinh(ke_1) \sinh(ke_2))} \end{aligned} \quad (1.77)$$

with $k = \sqrt{p/a_x}$.

Since both of these equations are explicit, it is easy to derive certain properties of a non-destructive thermal method. In such a method, measurement of the front or rear face thermal contrast time evolution (by infrared thermography for example) can produce some information about the parameters (position e_1 and resistance R_c) of the delamination.

Thus the position e_1 of the defect cannot be found by the rear face contrast measurement since e_1 and e_2 play symmetrical roles in equation (1.77): two delaminations, located for

example at one-quarter and three-quarters of the slab thickness, cannot be differentiated since they produce the same rear face contrast.

Equations (1.76) and (1.77) also show that, for low values of the contact resistance R_c , the second term in the denominator can be ignored, which makes the front and rear face contrasts proportional to this resistance. The reader might ask, at this point, how low this contact resistance should be, since it has a unit, to make this approximation valid. To answer this question, we must use a reduced Laplace transform of the reduced front or rear face temperature already met in Section 1.4.2:

$$\theta_{\alpha}^* = L^*[T_{\alpha}^*(t^*)] = \int_0^{\infty} T_{\alpha}^*(t^*) \exp(-p^* t^*) dt^* \quad \text{for } \alpha = f \text{ or } r \quad (1.78)$$

with

$$T_{\alpha}^*(t^*) = \rho c e T_{\alpha}(t)/Q; \quad t^* = at/e^2; \quad p^* = e^2 p/a$$

The reduced forms of equations (1.76) and (1.77) then become

$$\Delta\theta_f^* = \theta_f^* - \theta_{f0}^* = L^*[T_f^* - T_{f0}^*] = \frac{R_c^* \sinh^2(\alpha(1-z))}{\sinh(\alpha)(\sinh(\alpha) + \alpha R_c^* \sinh(\alpha z) \sinh(\alpha(1-z)))} \quad (1.79)$$

and

$$\Delta\theta_r^* = \theta_r^* - \theta_{r0}^* = L^*[T_r^* - T_{r0}^*] = \frac{R_c^* \sinh(\alpha z) \sinh(\alpha(1-z))}{\sinh(\alpha)(\sinh(\alpha) + \alpha R_c^* \sinh(\alpha z) \sinh(\alpha(1-z)))} \quad (1.80)$$

where

$$\alpha = \sqrt{p^*}; \quad R_c^* = R_c/(e/\lambda); \quad z = e_1/e$$

These last two equations show that the thermal contact is proportional to the contact resistance, as soon as its reduced value, R_c^* , is much smaller than unity, i.e. as soon as its value, R_c , is much smaller than the thermal resistance (e/λ) of the sane slab.

APPENDIX 1.1 MATLAB® CODE FOR THE HEAT PULSE ON AN INSULATED SLAB (See Section 1.4.1)

```
% Program Heat Pulse
% Thermophysical properties, thickness
lambda=2; a=1e-6; % Thermophysical properties, thickness
e=0.01 % Thickness (area S=1)
Q=1e5 % Energy density
% Coefficients of Stehfest's algorithm
```

```
v(1)=1/12;v(2)=-385/12;v(3)=1279;v(4)=-46871/3;v(5)=505465/6;
v(6)=-473915/2;v(7)=1127735/3;v(8)=-1020215/3;v(9)=328125/2;v(10)=-65625/2;
% loop on time
tv=0;Trear=0;Tfront=0;Tmid=0;Fmid=0;
for t=1:1:1000,
    p=(1:10)*log(2)/t;k=sqrt(p/a); % calculation of the Laplace parameters for a given time
    A=cosh(k*e);
    C=lambda*k.*sinh(k*e);
    Amid=cosh(k*e/2);
    Cmid=lambda*k.*sinh(k*e/2);
    Tetarear=Q./C;Tetafront=Q*A./C;Tetamid=Amid.*Tetarear;Phimid=Cmid.*Tetarear;
    tv=[tv t];
    rear=(log(2)/t).*(v.*Tetarear');Trear=[Trear rear];
    front=(log(2)/t).*(v.*Tetafront');Tfront=[Tfront front];
    midT=(log(2)/t).*(v.*Tetamid');Tmid=[Tmid midT];
    midF=(log(2)/t).*(v.*Phimid');Fmid=[Fmid midF];
end;
plot(tv,Trear,'k-',tv,Tmid,'k-',tv,Fmid/1000,'k-');
axis([-10 100 0 10]);
hold on
% pause
plot(tv,Tfront,'k-');
xlabel('Time (seconds)');
ylabel('Temperatures (Celsius), Flux (KW/m2)');

end
```

APPENDIX 1.2 MATLAB® CODE FOR THE FLUX STEP ON AN INSULATED SLAB (See Section 1.4.3)

```
% Program Flux step
% Thermophysical properties, thickness
lambda=2; a=1e-6; % Thermophysical properties, thickness
e=0.01 % Thickness (area S=1)
W=1e4 % Power density
roce=lambda*e/a % heat capacity per unit area of the slab
b=lambda/sqrt(a) % thermal effusivity
% Coefficients of Stehfest's algorithm
v(1)=1/12;v(2)=-385/12;v(3)=1279;v(4)=-46871/3;v(5)=505465/6;
v(6)=-473915/2;v(7)=1127735/3;v(8)=-1020215/3;v(9)=328125/2;v(10)=-65625/2;
% loop on time
tv=0;T=0;
for t=1:1:200;
    p=(1:10)*log(2)/t;k=sqrt(p/a); % calculation of the Laplace parameters for a given time
    A=cosh(k*e);
    C=lambda*k.*sinh(k*e);
    Teta=(W./p)/A./C;
    tv=[tv t]; % vector of the different times the solution is calculated
    front=(log(2)/t).*(v.*Teta');
```

```

T=[T front]; % front face temperature
end;
Tshort=(W/roce)*tv; % short time solution
Tlong=(2*W/b/sqrt(pi))*sqrt(tv); % long time solution
plot(tv,T,'k',tv,Tshort,'k',tv,Tlong,'k');
hold on
end

```

APPENDIX 1.3 MATLAB® CODE FOR A TWO-LAYER BUILDING WALL IN A TRANSIENT TRANSFER (See Section 1.4.4)

```

% Program Two-layerWall
clear;
e1=0.20,e2=0.05 % thicknesses (area S=1)
lbd1=1.7,lbd2=0.16, % conductivities
a1=1.2e-6,a2=1.6e-7 % diffusivities
TH=15,T0=0 % temperatures
hins=10,hout=10 % heat transfer coefficients
% Coefficients of Stehfest's algorithm
v(1)=1/12;v(2)=-385/12;v(3)=1279;v(4)=-46871/3;v(5)=505465/6;
v(6)=-473915/2;v(7)=1127735/3;v(8)=-1020215/3;v(9)=328125/2;v(10)=-65625/2;
% loop on time
tv=0;T1=0;T2=0;
for t=360:360:360000,
    p=(1:10)*log(2)/t;k1=sqrt(p/a1);k2=sqrt(p/a2);
    % calculation of the Laplace parameters for a given time
    A1=cosh(k1*e1);A2=cosh(k2*e2);
    B1=(1./(lbd1*k1)).*sinh(k1*e1);B2=(1./(lbd2*k2)).*sinh(k2*e2);
    C1=lbd1*k1.*sinh(k1*e1);C2=lbd2*k2.*sinh(k2*e2);
    A=A1.*A2+B1.*C2;B=A1.*B2+A2.*B1;C=C1.*A2+A1.*C2;D=C1.*B2+A1.*A2; % coefficients of
    the wall matrix
    Bc=A/hout+B+C/(hout*hins)+D/hins;
    Dc=C/hout+D;
    Phiout=(TH-T0)/p./Bc; % Laplace flux at the outside face
    Teta1=(A/hout+B).*Phiout; % Laplace temperature at the inside face
    Tetai=(A2/hout+B2).*Phiout; % interface Laplace temperature
    Teta2=Phiout/hout; % Laplace temperature at the outside face
    tv=[tv t]; % vector of the different times for which the solution is calculated
    T1=[T1(log(2)/t).*(v*Teta1)]; % inside face temperature
    Ti=[Ti(log(2)/t).*(v*Tetai)]; % interface temperature
    T2=[T2(log(2)/t).*(v*Teta2)]; % outside face temperature
end;
    tv=tv/3600; % time in hours
    plot(tv,T1,'k',tv,Ti,'k',tv,T2,'k');
    axis([-10 100 0 15]);
    xlabel('Time (hours)');
    ylabel('Temperatures (Celsius)');
end

```

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